Problem

Consider a graph G with edges E and vertices V, and subgraphs $G_1, \ldots, G_k \subset G$. Let $G_i = (E_i, V_i)$ for these subgraphs. For now let us suppose the graph is unweighted and undirected. The subgraphs need not cover G and can overlap. Each player i knows G and G_i, but not the other player's graphs. (This is akin to a map of a city, where each player is a different regional taxi company.) The goal is, given some starting vertex $v \in V$, and some end vertex, $w \in V$ to:

1. Identify if it is possible to construct a path $v, v_1, \ldots, v_l, w$ such that every edge on the path is in $\bigcup_{i=0}^{k} E_i$.
2. If such a path exists, find the minimum number of edges needed to get from v to w, while using only edges belonging to $\bigcup_{i=0}^{k} E_i$.

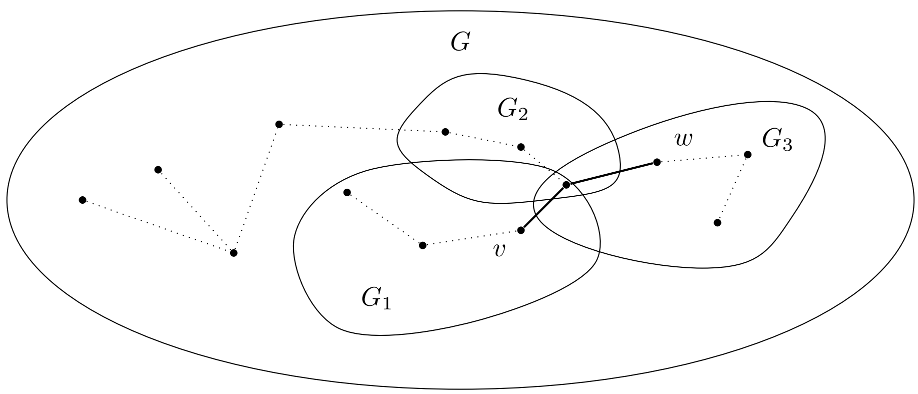

Figure 1: A valid path from v to w in bold. Note that G may contain nodes not belonging to any player.

Lower bound sketches and ideas

For **??**, we think this would require computing $\binom{k}{2}$ pairwise set disjointnes in the worst case, to determine where the intersections of each G_i are, at which point each player could on their own determine the possible paths between each of the subgraphs that intersect with their own.

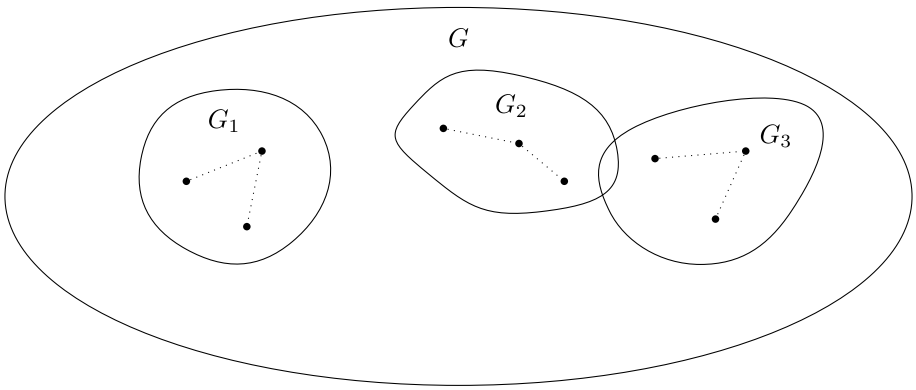

Figure 2: Worst case for **??**: Disjoint Subsets

For **??**, we would be required to compute $\binom{k}{2}$ pairwise intersections, since in order to minimize the path we would need to know all possible paths that could connect v and w. To find the shortest path, each player works backwards and sends the distances through their internal vertices to the players they intersect with, and each player will attempt to identify if a longer path can be dropped whenever they receive this list.

Lower bound using Equality

Equality problem. In the two-party Equality problem, Alice is given a string $x \in \{0,1\}^n$ and Bob is given a string $y \in \{0,1\}^n$. The goal is to determine whether $x = y$. It is known that Equality has deterministic communication complexity $\Omega(n)$.

Equality gadget. We construct a gadget that enforces equality at a single bit position. The gadget is shown in Figure **??**. The gadget consists of two parallel branches.

1. The top branch corresponds to the assignment $x_i = y_i = 1$,
2. The bottom branch corresponds to the assignment $x_i = y_i = 0$.

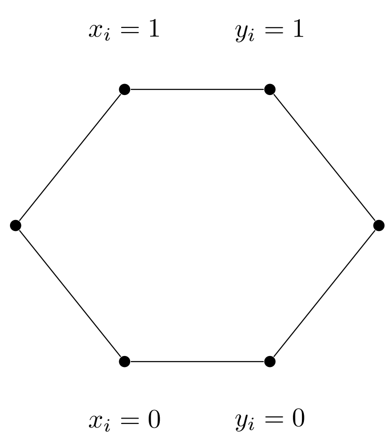

Figure 3: Gadget for proving equality. A path only exists if $x_i = y_i$ for either branch.

Alice enables exactly one entry node of the gadget depending on x_i, and Bob enables exactly one exit node depending on y_i. A path exists through the gadget if and only if Alice and Bob select nodes on the same branch, which holds exactly when $x_i = y_i$.

Reduction to our graph problem. We construct a graph by chaining one copy of this gadget for each index $i \in [n]$ between a global start vertex v and end vertex w. In the resulting graph, there exists a path from v to w if and only if all gadgets are traversable, which occurs if and only if $x_i = y_i$ for all i, i.e., if and only if $x = y$.

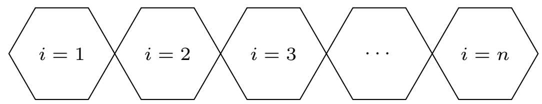

Figure 4: Equality gadgets chained as adjacent hexagonal modules.

Therefore, any protocol that solves the reachability problem on this graph can be used to solve the Equality problem. Since Equality requires $\Omega(n)$ deterministic communication, it follows that our graph problem also requires $\Omega(n)$ deterministic communication.

Lower bound using Disjointness

Set Disjointness problem. Alice is given a vector $x \in \{0,1\}^n$ and Bob is given a vector $y \in \{0,1\}^n$. The goal is to determine whether the sets represented by x and y are disjoint. It is known that Disjointness has deterministic communication complexity $\Omega(n)$.

Disjointness gadget. We construct a gadget that enforces disjointness at a single bit position. The gadget is shown in Figure 5. The gadget is constructed so that traversal is possible for all assignments except when both parties contain the element.

1. The top branch corresponds to when x_i has the element but y_i does not
2. The middle branch corresponds to when x_i and y_i do not have the element
3. The top branch corresponds to when y_i has the element but x_i does not

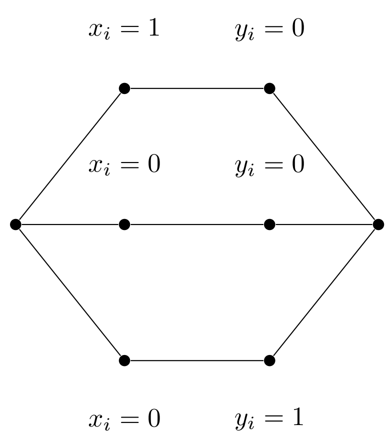

Figure 5: Gadget for proving equality, a path only exists if x_i and y_i do not have the same element

Reduction to our graph problem. We construct a graph by chaining one copy of this gadget for each index $i \in [n]$ between a global start vertex v and end vertex w. In the resulting graph, there exists a path from v to w if and only if all gadgets are traversable, which occurs if and only if x and y does not contain the same element

Therefore, any protocol that solves the reachability problem on this graph can be used to solve the Disjoint problem. Since Disjoint requires $\Omega(n)$ deterministic

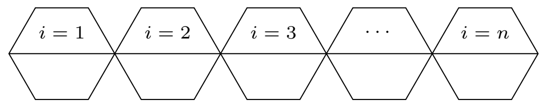

Figure 6: Equality gadgets chained as adjacent hexagonal modules.

communication, it follows that our graph problem also requires $\Omega(n)$ deterministic communication.